Natural Language Processing Coursework

Name:	Khalid Alahmadi
CID:	*06051115*
Leaderboard name:	`kaa525`

Repository link:
https://github.com/kaa25-ic/NLP_CW

1 Exercise 1: Critical Paper Review

Primary contributions:

1. **A new dataset and taxonomy for PCL towards vulnerable communities.** The authors introduce the *Don't Patronize Me!* dataset comprising 10,637 English-language news paragraphs spanning 20 countries and multiple vulnerability-related keywords, explicitly targeting *patronising and condescending language (PCL)* directed at vulnerable groups (Pérez-Almendros et al. [2020]). Crucially, they also propose a structured two-level taxonomy of PCL with three high-level families (*The Saviour, The Expert, The Poet*) and seven fine-grained categories (e.g., unbalanced power relations, shallow solutions, presuppositions, authority voice, compassion, metaphor, and "the poorer, the merrier"). This combination provides both a clear conceptualisation of PCL and a practical foundation for developing and evaluating NLP systems.

2. **Benchmarking and analysis that highlight the technical challenges of PCL detection.** The paper establishes baseline performance for (i) paragraph-level binary PCL detection and (ii) paragraph-level multi-label category prediction, comparing classical approaches (SVM with BoW/Word2Vec), a BiLSTM, and several transformer models (BERT variants, RoBERTa, DistilBERT). Beyond reporting precision/recall/F1, the authors analyse per-category difficulty and provide qualitative error examples, showing that while some categories correlate with relatively surface cues (e.g., compassion-related phrasing), others require non-trivial world knowledge (e.g., judging whether a proposed "solution" is shallow or whether a presupposition is warranted). These baselines and analyses set a strong reference point for subsequent work and motivate methodological improvements.

Technical strengths that justify publication:

1. **Methodologically careful resource creation with a clear, socially grounded problem framing.** The work addresses a form of harmful language that is subtle, often unconscious, and distinct from overt hate speech, making it both socially relevant and technically non-trivial. The dataset is produced via a structured two-step annotation pipeline (paragraph-level identification followed by span/category annotation) with expert annotators; the authors report inter-annotator agreement and describe mechanisms for resolving disagreements (including a referee annotator), increasing transparency and credibility.

2. **Strong experimental baselines and insightful analysis.** The paper evaluates diverse model families under a consistent protocol and reports appropriate metrics (precision/recall/F1 for the positive PCL class), complemented by per-category results and qualitative error analysis. This empirical breadth makes the dataset immediately usable by the community and provides concrete evidence of where current models succeed or fail, strengthening the paper's scientific value.

Key weaknesses :

1. **Annotation subjectivity and moderate agreement.** Paragraph-level PCL identification yields only moderate inter-annotator agreement when considering all labels, indicating that the task is inherently subjective and that labels may contain substantial noise. While borderline cases are explicitly modelled via a multi-point scale, the paper could further characterise

disagreement patterns (e.g., by category, keyword, or country) to better justify label reliability and guide modelling choices.

2. **Limited discussion of class imbalance and its implications.** The dataset contains relatively few positive PCL instances compared to negatives, which can make F1 highly sensitive to thresholding and training dynamics. However, the paper provides limited analysis of imbalance-aware strategies (e.g., reweighting, sampling, or calibration), and does not explore how such decisions affect error types (false positives vs. false negatives) across categories.

2 Exercise 2: Exploratory Data Analysis

We report four distinct EDA techniques. For each, we provide evidence, analysis, and an impact statement describing how the insight affected modelling decisions. A detailed notebook "NLP_CW_EDA" is included as a comprehensive reference for below results.

2.1 Basic Statistical Profiling

Visual/Tabular Evidence. Table 1 summarises the class imbalance in Train/Dev. Table 2 reports text-length statistics (words, characters), and Table 3 summarises RoBERTa token-length and truncation-related evidence. Figure 1 and Figure 2 provide visual confirmation.

Split	Total	Non-PCL (0)	PCL (1)	PCL Rate (%)	Imbalance (0:1)
Train	8,375	7,581	794	9.48	9.55:1
Dev	2,094	1,895	199	9.50	9.52:1

Table 1: Class distribution in Train/Dev (strong imbalance; splits are well matched).

Split	Metric	Min	Mean	Median	Max
Train	Word length	1	48.68	40	909
Train	Char length	7	260.26	220	4942
Dev	Word length	0	47.41	40	272
Dev	Char length	0	251.80	214	1571

Table 2: Text-length statistics (words/chars). Dev contains at least one empty/whitespace paragraph (min=0).

Split	Token metric (RoBERTa)	Mean	Median	P95	Max
Train	Token length	55.91	49	115	1005
Dev	Token length	54.54	48	112	355

Table 3: RoBERTa token-length profile. Most examples fall below typical max_length settings, but there is a long tail of outliers.

Analysis. The dataset is strongly imbalanced with only $\approx 9.5\%$ PCL examples, but Train and Dev have nearly identical class proportions (Table 1), supporting reliable evaluation. Text lengths are typically moderate (Table 2), but there are extreme outliers and at least one empty dev example. The token-length distribution (Table 3, Figure 2) shows that most samples fit comfortably under standard Transformer limits, while a small tail will be truncated.

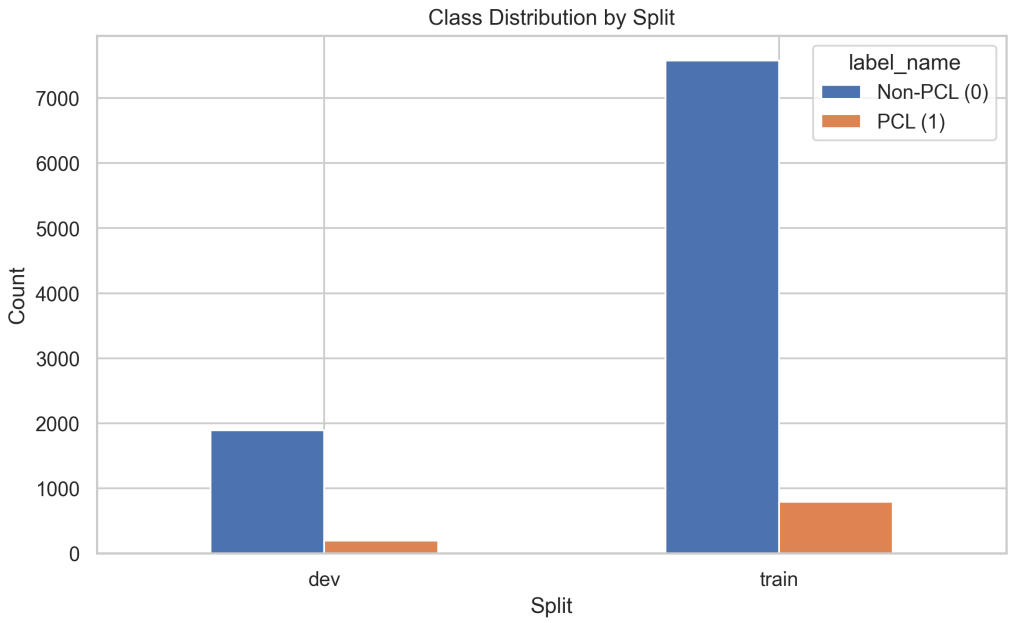

Figure 1: Class distribution bar plot for Train and Dev.

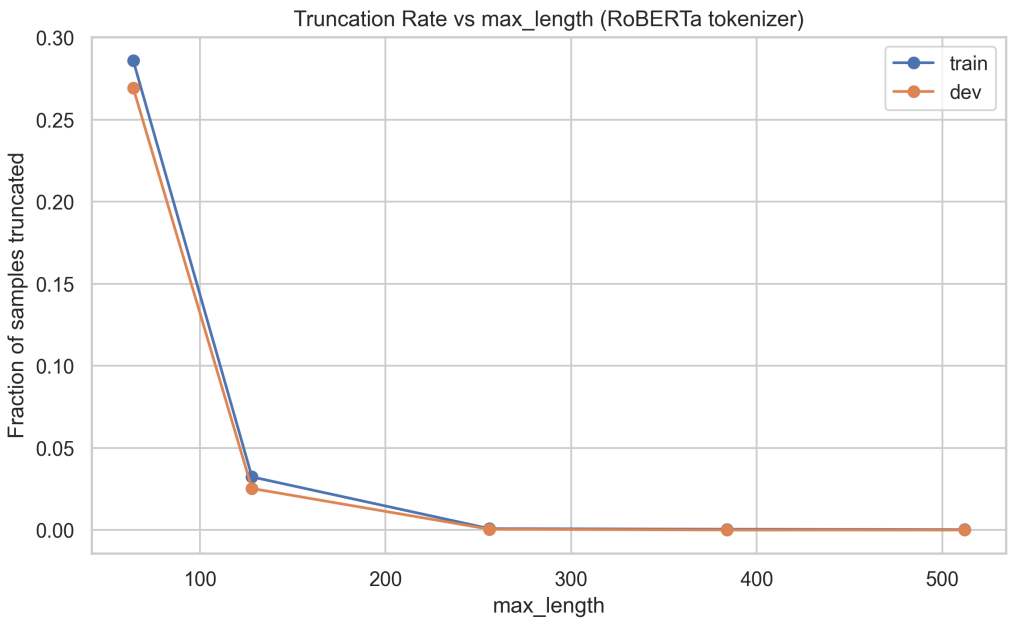

Figure 2: Histogram of RoBERTa token lengths (Train/Dev), highlighting the long tail that may be truncated under fixed max_length.

Impact Statement. Given the ∼9.5:1 class imbalance, accuracy is not a sufficient evaluation metric; the approach should prioritise positive-class performance (e.g., F1) and consider imbalance-aware strategies (class weighting, threshold tuning). The token-length analysis motivates a justified `max_length` choice paired with padding for efficiency, while long-tail outliers and empty/whitespace paragraphs should be handled explicitly to avoid instability during training/evaluation.

2.2 Lexical Analysis

Visual/Tabular Evidence. Table 4 lists representative top unigrams by class (stopwords removed). Table 5 provides examples of over-represented tokens by class (document-frequency ratio). Table 6 shows stopword-density differences.

Class	Representative top unigrams (stopwords removed)
PCL	people, need, poor, children, families, homeless, women, help, disabled, refugees
Non-PCL	people, women, immigrants, families, vulnerable, need, disabled, poor, refugees, said

Table 4: Representative frequent unigrams by class.

Direction	Examples of over-represented tokens
PCL-leaning	hungry, compassion, christmas, hearts, donate, underprivileged
Non-PCL-leaning	anti, released, fund, increase, serious

Table 5: Examples of tokens over-represented by class using a document-frequency ratio heuristic.

Split	Non-PCL mean	PCL mean
Train	0.3956	0.4287
Dev	0.3948	0.4282

Table 6: Mean stopword density by class. PCL exhibits higher stopword density in both Train and Dev.

Analysis. Lexical content overlaps substantially across classes (e.g., references to *poor*, *families*, and target groups appear frequently in both), suggesting that topic alone is insufficient for PCL detection. In contrast, over-represented PCL tokens often reflect stance/framing (e.g., charity and sympathy cues such as *donate* or *compassion*). PCL samples also have higher stopword density (Table 6), consistent with more narrative or connective phrasing.

Impact Statement. Because keywords and topical references appear in both classes, a contextual model (e.g., Transformer) is better suited than bag-of-words approaches: it can capture patronising *framing* and tone rather than mere topic. The stopword-density gap suggests that stylistic cues (function-word patterns, narrative phrasing) may contribute to separability, motivating careful threshold selection and qualitative error analysis around charity/sympathy framing.

2.3 Semantic & Syntactic Exploration

Visual/Tabular Evidence. Table 7 summarises mean POS proportions by class, and Table 8 summarises entity counts/density by class. Figures 3 and 4 show the full distribution of POS and NER for the training data set.

Split	POS feature	Non-PCL mean	PCL mean
Train	Pronouns (PRON)	0.0570	0.0789
Train	Verbs (VERB)	0.1200	0.1300
Train	Proper nouns (PROPN)	0.0917	0.0755
Dev	Pronouns (PRON)	0.0550	0.0811
Dev	Verbs (VERB)	0.1202	0.1309
Dev	Proper nouns (PROPN)	0.0917	0.0768

Table 7: Selected POS proportion means by class. PCL shows higher pronoun usage and lower proper-noun usage across splits.

Split	Entity feature	Non-PCL mean	PCL mean
Train	Entity count	3.4648	2.7620
Dev	Entity count	3.4923	2.6030
Dev	Entity density	0.0872	0.0538

Table 8: Named-entity evidence: non-PCL tends to contain more entities (count and density), consistent with a more factual/news-like style.

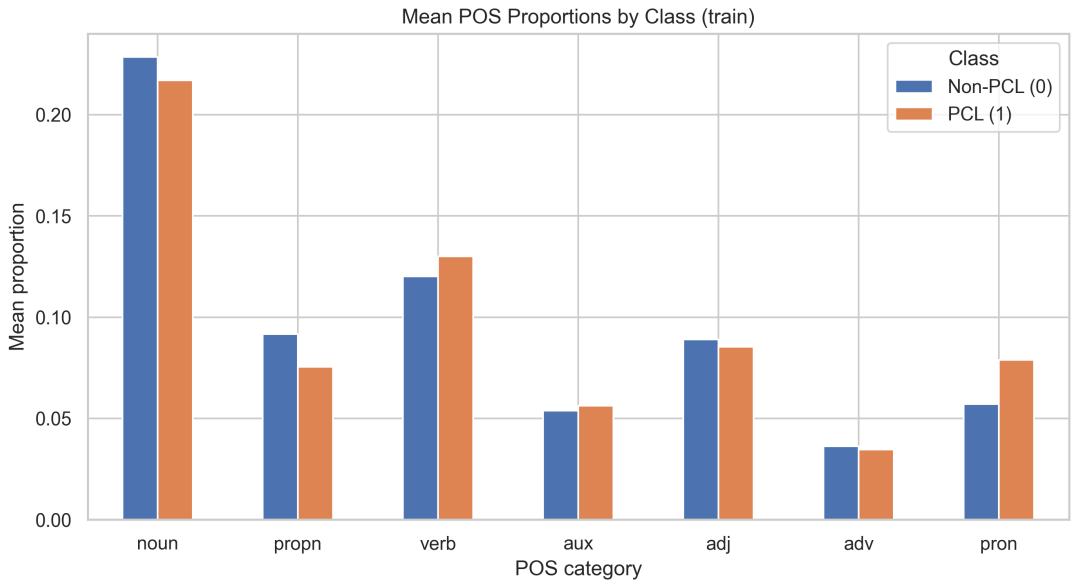

Figure 3: POS distribution comparison across classes (Train set).

Analysis. PCL text contains a higher proportion of pronouns and slightly more verbs, suggesting more direct-address or narrative framing. Conversely, non-PCL text contains more named entities (Table 8), consistent with more factual reporting that references specific organisations/places/people. Together, these patterns indicate that PCL is primarily a discourse/style phenomenon rather than a topic signal.

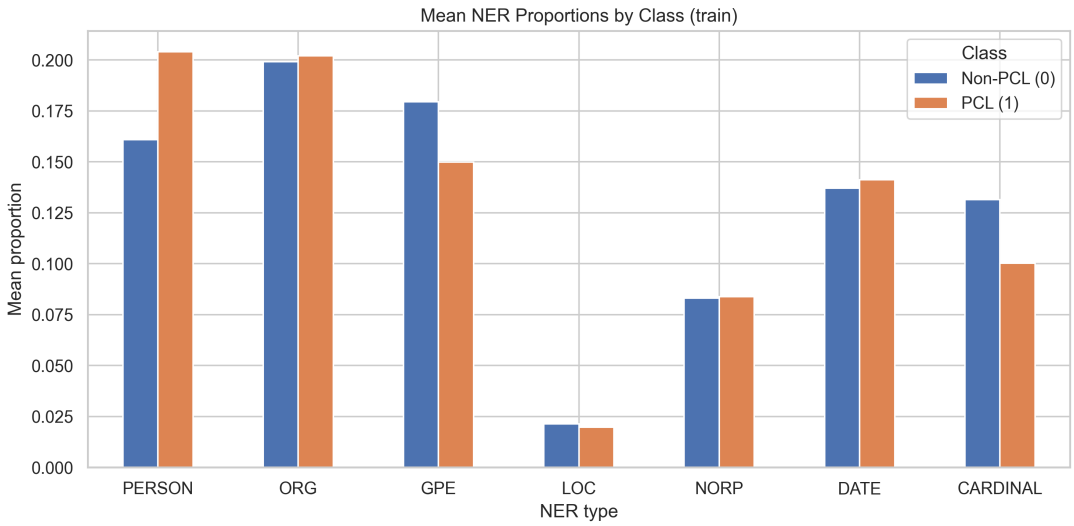

Figure 4: NER counts/density comparison across classes (Train/Dev).

Impact Statement. These findings support modelling choices that capture framing and discourse (contextual encoders) rather than relying on entity or topic related features.

2.4 Noise & Artifact Detection

Visual/Tabular Evidence. Table 9 summarises dataset artifacts (HTML tags/entities, quotation prevalence, extreme-length rates) and data-quality checks (empties and duplicates).

Check / Artifact	Dev (%)	Train (%)
Empty/whitespace-only texts	0.05	0.00
Within-split exact duplicates	0.00	0.00
Train–Dev exact overlap	0.00	0.00
URLs present	0.10	0.08
HTML tags present	4.06	3.77
HTML entities present	0.62	0.78
Multiple quotes present	24.07	25.01
Repeated punctuation present	2.24	2.67
Very short texts	0.10	0.06
Very long texts	1.91	2.05
Punctuation-heavy texts	5.30	4.96

Table 9: Noise and artifact summary. Quote-heavy text is common; HTML artifacts appear in a small but non-trivial fraction; no duplicate leakage was detected.

Analysis. Overall, the dataset appears clean with no evidence of train–dev leakage via exact duplicates. However, several artifacts merit attention: HTML tags/entities appear in a minority of samples, and quote-heavy formatting is very common (around one quarter of the data). Additionally, there is at least one empty dev sample, and a small fraction of texts are extreme in length or punctuation density.

Impact Statement. Robust preprocessing is required to prevent spurious cues and evaluation instability. Preprocessing may include handling empty/whitespace samples, normalising/stripping HTML artifacts, and standardising quotation patterns.

3 Exercise 3: Proposed Approach

3.1 Overview of Our Approach

Similarly to the SemEval baseline model, we use **RoBERTa-base**, but our implementation further improves PCL classification through: (i) **conservative text normalisation** to reduce dataset artifacts, (ii) **systematic hyperparameter tuning**, and (iii) a set of **controlled experiments** using class weighting, oversampling and thresholding to handle class imbalance, selecting the best-performing configuration.

3.2 Model Choice: RoBERTa-base

We fine-tune **RoBERTa-base** (Robustly Optimized BERT Pretraining Approach), a Transformer encoder pretrained with masked language modelling on large corpora (Liu et al. [2019]). RoBERTa-base is appropriate for PCL detection because PCL is typically conveyed via **subtle framing, stance, and discourse cues** (e.g., condescending tone, implied power asymmetry), which benefit from contextual representations. In contrast to keyword-based methods, RoBERTa can model how vulnerable groups are *described* rather than merely whether they are mentioned. Additionally, this model performed the best in the original "Don't Patronize Me!" paper (Pérez-Almendros et al. [2020]).

3.3 Data Split and Reproducibility

We reconstruct the official dataset and respect the official Train/Dev split by paragraph IDs as proposed by SemEval2022 Task 4 (Pérez-Almendros et al. [2022]). The official development set is treated as a final evaluation set in the coursework pipeline. For model selection, we further create a stratified 80/20 Train/Validation split from the official training data to tune modelling decisions while maintaining label proportions.

All experiments fix random seeds across Python/NumPy/PyTorch and record the full configuration (hyperparameters, preprocessing, and training settings) to ensure reproducibility.

Finally, once the best model setup is determined, training and validation sets are combined for one final model training on the entire data set to benefit from the additional training samples and to boost the F1 score.

3.4 Preprocessing

The corpus contains formatting artifacts (e.g., HTML entities, inconsistent punctuation and whitespace). We apply **light normalisation** that removes noise while preserving semantic and stylistic cues relevant to PCL. We produce cleaned text via:

- **HTML entity unescaping** (e.g., `&` → `&`, `"` → `"`).
- **Whitespace normalisation**: replace explicit newline markers (e.g., `\n`) and real line breaks/tabs with spaces; collapse repeated whitespace.
- **Unicode punctuation normalisation**: map curly quotes and long dashes to ASCII equivalents.
- **Contraction repair**: fix spaced contractions (e.g., `do n ' t` → `don't`).

This choice is intentionally conservative: it reduces spurious tokenisation patterns (noise) without stripping punctuation or stylistic signals (potentially informative for condescension).

3.5 Tokenisation

We tokenise using the **RoBERTa-base tokenizer** with truncation and padding:

$$\texttt{truncation=True}, \quad \texttt{padding=max_length}, \quad \texttt{max_length} = 256.$$

A maximum length of 256 tokens balances retaining most paragraphs while keeping fine-tuning computationally efficient.

3.6 Training Setup

We fine-tune a 2-label sequence classification head on top of RoBERTa-base using the Hugging Face training pipeline (Wolf et al. [2020]). We use AdamW optimisation with weight decay and evaluate each epoch. Model selection is performed using **validation F1** (positive class), which is more informative than accuracy under class imbalance.

We report accuracy and positive-class precision/recall/F1:

$$\mathrm{F1} = \frac{2PR}{P+R}, \qquad P = \frac{TP}{TP+FP}, \qquad R = \frac{TP}{TP+FN}.$$

3.7 Experiments and Improvements over the Shared-task Baseline

To surpass the baseline, we conducted controlled experiments around (i) training stability, (ii) class imbalance, and (iii) calibration. We start with the preprocessed data and the basic training setup described earlier, then introduce thresholding, weighted loss, oversampling, and finally a hyperparameter sweep.

Threshold Tuning After each training run, we obtain the model's positive-class probabilities on the validation set:

$$p_i = p(y{=}1 \mid x_i), \quad i = 1, \ldots, n.$$

We then sweep a grid of thresholds $t \in \mathcal{T}$ and convert probabilities into hard predictions:

$$\hat{y}_i(t) = \mathbb{I}[p_i \geq t].$$

For each threshold, we compute precision, recall, and F1 for the positive class, then the selected threshold is:

$$t^\star = \arg\max_{t \in \mathcal{T}} \mathrm{F1}(t).$$

In our experiments, we use a dense grid (e.g., $t \in \{0.01, 0.02, \ldots, 0.99\}$) and also inspect coarser grids for stability.

Although our final submitted model does *not* use threshold tuning, we evaluate it as a calibration method for improving positive-class F1 under class imbalance. Threshold tuning did sometimes improve validation F1, but the optimal $t^\star$ was sensitive and produced noisy behaviour across runs. For robustness and reproducibility, we therefore retained the default decision rule (threshold 0.5) in the final model.

Class-weighted Training To mitigate the strong class imbalance, we tested a weighted cross-entropy loss where errors on the minority class (PCL) are penalised more heavily.

A **9.5:1** emphasis toward the minority class was used.[1] We also experimented with moderated weight ratios (e.g., clipping w_1 to reduce over-correction), but ultimately did not include class weighting in the final submission since our best-performing configuration was achieved via hyperparameter tuning without sample weighting.

Oversampling the Minority Class We also evaluated naive oversampling as a data-level approach to address imbalance. Specifically, we replicate PCL training instances by a fixed factor k (e.g., $k = 5$) and shuffle the resulting training set. Oversampling increases exposure to minority patterns, but can increase overfitting; therefore it was treated as an ablation and compared against the unmodified training distribution.

Hyperparameter Sweep Our strongest gains came from a structured sweep over fine-tuning hyperparameters. For each configuration, we trained on the training split and evaluated on the validation split, selecting the best model by validation F1 (positive class).

Sweep configuration. We explored the following grid:

- Learning rate: $\{1 \times 10^{-5}, 2 \times 10^{-5}, 3 \times 10^{-5}\}$
- Epochs: $\{3, 4\}$
- Batch size: $\{8, 16\}$
- Weight decay: $\{0, 0.01\}$

All runs used the same preprocessing pipeline and tokenisation settings (`max_length`=256) to ensure comparability. Where applicable, we used early stopping (patience=2) based on validation F1 to avoid overfitting, and we retained the checkpoint with the best validation F1.

3.8 Rationale and Expected Outcome

Our approach is expected to outperform the shared-task baseline because it improves generalisation through a cleaner input representation (artifact normalisation) and a carefully tuned fine-tuning regime. Since PCL signals are subtle and heavily dependent on phrasing and context, these changes help the model learn more robust representations of condescending framing while avoiding over-reliance on noisy formatting cues.

3.9 Final Configuration (BestModel)

Table 10 reports the final configuration used in our best-performing model.

4 Exercises 4 and 5.1: Modelling

We generated and pushed two prediction files to our repository: `dev.txt` (predictions for the official development set) and `test.txt` (predictions for the official test set). Each file contains binary labels, while preserving the original ordering of the provided datasets.

[1] Using the dataset counts observed in EDA, $N_0/N_1 \approx 9.5$.

Component	Setting
Backbone	`roberta-base`
Max length	256
Training objective	Cross-entropy loss
Optimiser	AdamW
Learning rate	`1e-5`
Batch size	`16`
Epochs	`4`
Weight decay	`0.01`
Early stopping	`True`
Decision rule	Default (no threshold tuning)

Table 10: Final training configuration.

All predictions were produced using our **BestModel**, a fine-tuned **RoBERTa-base** trained with conservative text normalisation (e.g., HTML/whitespace/punctuation clean-up) and a hyperparameter-tuning as mentioned in Table 10. This model achieved a best positive-class **F1 score of 0.579**.

4.1 Exercise 5.2: Local Evaluation

4.1.1 Ablation Study

To understand which components of our approach contribute most to performance, we conducted an ablation study. Starting from a baseline RoBERTa-base setup, we introduced one modification at a time (e.g., preprocessing, thresholding, imbalance-handling strategies), and compared the resulting positive-class F1 scores on the validation set (except for the BestModel which was scored on the official development set). Table 11 summarises the experiments and their outcomes.

Experiment	F1 (Val)	Remarks
Baseline	0.545	No preprocessing.
Preprocessing	0.544	Light preprocessing.
Thresholding	0.566	Preprocessing and a decision threshold of 0.05.
Sample weighting	0.543	Preprocessing and class-weighted loss with a ratio of 9.5:1 (PCL:No-PCL).
Oversampling	0.548	Preprocessing and oversampling factor of 5 for the PCL class.
BestModel	0.579 (Dev)	Preprocessing + hyperparameter sweep + retraining on the combined training and validation set.

Table 11: Ablation study results (validation split). F1 is reported for the positive class (PCL).

The ablation results should be interpreted in light of the dataset characteristics: the positive class (PCL) is rare (approximately 9.5%), so our validation split contains relatively few PCL instances. Consequently, small fluctuations in F1 (e.g., ± 0.001–0.005) can arise and should be treated as *noise-level* differences rather than definitive improvements.

Preprocessing yields an essentially unchanged F1 (0.545 to 0.544), suggesting that RoBERTa is already robust to minor textual artifacts and that our normalisation is conservative. **Sample weighting** slightly reduces F1 (to 0.543), which is consistent with a common trade-off in imbalanced classification: increasing the penalty on PCL can raise recall but may also increase false positives, lowering precision and hence F1 when over-correction occurs. Additionally, Transformers already learn enough minority signal, and hence weighting can shift calibration and hurt precision under extreme imbalance. **Oversampling** provides a modest gain (to 0.548),

plausibly due to increased exposure to minority patterns, although the effect remains negligible. We decided to only include preprocessing in the final model as it did not hurt performance and seemed logical to include, while both sample weighting and oversampling were excluded.

Threshold tuning yields the largest gain among the single-step modifications (F1 = 0.566) by moving the decision boundary to better optimise the positive-class F1 objective under severe class imbalance. In practice, however, the threshold that maximised F1 varied noticeably across runs, producing noisy and non-reproducible behaviour. We also found that different thresholds (e.g., 0.05 and 0.18) could achieve similar F1 scores via different precision–recall trade-offs, while very small thresholds could substantially distort either precision or recall. For robustness and reproducibility, we therefore did not apply threshold tuning in the final submission. Finally, our **BestModel** attains the highest overall score (F1 = 0.579), driven primarily by an improved fine-tuning configuration identified through hyperparameter search and subsequent retraining on the combined training and validation data. Because **BestModel** is evaluated on the official development set, it is not strictly comparable to the validation-only ablations, but it represents the strongest configuration selected for global evaluation.

4.1.2 Error Analysis and Comparison with the Baseline

In this section, we conduct local evaluation via error analysis to better understand the model's behaviour and failure modes. Specifically, we compare (i) the **Baseline** RoBERTa-base configuration from Table 11 and (ii) our final **BestModel**. For a fair comparison, we evaluate *both* models on the **official development set** (baseline model was reevaluated on the official development set, hence the different F1 score compared to Table 11). All results are available in the notebook "NLP_CW_Evaluation".

Overall behaviour and confusion analysis. BestModel improves positive-class F1 on the official dev set by approximately 2%, driven mainly by a precision gain (0.557 to 0.640) at the cost of lower recall (0.563 to 0.528). Table 12 shows the confusion matrices for both models. BestModel reduces false positives (89 to 59), indicating a more conservative decision boundary, but increases false negatives (87 to 94), meaning it misses some PCL examples that the baseline captures.

Model	TP	FP	FN	TN	Accuracy	Precision	Recall	F1
Baseline	113	89	87	1905	0.916	0.557	0.563	0.560
BestModel	106	59	94	1935	0.927	0.640	0.528	0.579

Table 12: Confusion matrix counts on the official dev set.

Ordinal label (`orig_label`) and predicted PCL rate. To understand whether errors cluster in borderline vs clear cases, we computed the predicted PCL rate for each ordinal label group proposed in the original paper (Table 13). The true PCL rate is 0 for labels 0–1 and 1 for labels 2–4. BestModel predicts PCL far less often for `orig_label`=0 and 1 than the baseline, consistent with fewer false positives. For positive groups, BestModel's predicted PCL rate is comparable for `orig_label`=4 but lower for `orig_label`=3, suggesting that moderately positive PCL (label 3) remains harder than strongly positive PCL (label 4).

Performance with the 7 PCL types. Although our model is trained for binary classification, we use the 7 PCL types reported in the original paper for diagnosis by computing subtype-specific

orig_label	Count	Gold PCL rate	Baseline PCL rate	BestModel PCL rate
0	1704	0.0000	0.0288	0.0188
1	191	0.0000	0.2094	0.1414
2	18	1.0000	0.2222	0.2222
3	89	1.0000	0.4944	0.4045
4	92	1.0000	0.6957	0.7065

Table 13: Gold PCL rate and predicted PCL rates by ordinal label group on the official dev set.

false negative (FN) rates (Table 14) (Pérez-Almendros et al. [2020]). BestModel improves on **Shallow solution** (FN rate 0.389 to 0.333), matches the baseline on **Authority voice** and **The poorer, the merrier**, but has higher FN rates on **Presupposition**, **Compassion**, **Metaphor**, and **Unbalanced power relations**. This explains why the overall improvement is precision-driven: BestModel reduces over-prediction on negatives, but still misses several positive instances. One possible explanation for the improvement in detecting "shallow solutions" is that it often includes charity/solution framing that may be easier to pick up lexically, while presupposition/metaphor requires more world knowledge and pragmatic inference.

PCL Type	# of examples	Baseline FN rate	BestModel FN rate	Δ (Best–Base)
Presupposition	62	0.500	0.581	+0.081
Compassion	106	0.377	0.434	+0.057
Metaphor	52	0.365	0.423	+0.058
Unbalanced power relations	82	0.366	0.401	+0.035
Shallow solution	36	0.389	0.333	−0.056
Authority voice	38	0.447	0.447	+0.000
The poorer, the merrier	11	0.636	0.636	+0.000

Table 14: Subtype-specific false negative (FN) rates on the official dev set.

Qualitative inspection of BestModel decisions. To complement the quantitative trends above, we manually inspected representative development-set instances from four diagnostic groups: (i) *positives fixed by BestModel* (cases where the baseline missed PCL but BestModel correctly predicts PCL), (ii) *regressions introduced by BestModel* (baseline correct but BestModel incorrect), (iii) *BestModel false positives* (non-PCL flagged as PCL), and (iv) *BestModel false negatives* (PCL missed by BestModel). As shown in Table 15, corrected positives often contain explicit patronising framings (e.g., exaggerated compassion, prescriptive "solution" language, or stereotyping), suggesting that BestModel is better at capturing surface and discourse markers aligned with PCL. In contrast, regressions and false positives frequently involve emotionally charged but non-patronising content or ambiguous rhetorical style, where the model appears to over-trigger on sympathy-related vocabulary. Finally, the false negatives in Table 15 tend to rely on subtler pragmatic signals (e.g., presuppositions or metaphorical framing) that require wider context or world knowledge, consistent with the higher miss rates observed for these subtypes in the quantitative analysis. Overall, the examples in Table 15 provide qualitative evidence that BestModel's gains are largely driven by improved sensitivity to common PCL cues, while remaining errors concentrate around ambiguous affective language and implicit, context-dependent patronisation.

Conclusion BestModel's error pattern suggests it is optimised for avoiding spurious positives: it predicts PCL relatively rarely on clear negatives, but still struggles with (i) **implicit** conde-

Group	par_id	Gold	Base	Best	Text
(i)	5612	1	0	1	”He said the victims who are currently rendered homeless can now be relieved of troubles as the 5,000 iron sheets from Mwanza had arrived , with 1,200 already distributed to victims in Bukoba Municipality” **(Shallow solution)**
(i)	10269	1	0	1	“Veterans left on scrapheap : The homeless plight of former servicemen and women” **(Metaphor/Compassion framing)**
(ii)	4580	1	1	0	“Old BC : As usual , Old GB has it assy-versy . I think there should be more focus on preventing homelessness than curing it . Are there systems for actually going out , picking up the homeless , and placing them where they belong ? Homeless people are n’t voluntarily walking into these shelters” **(subtle framing)**
(ii)	10253	1	1	0	”Rather sad. Good set of pictures that tells a tale of survival , subsistence living , and hopeless nature of life in the tribal societies . Exploiting an unexpected geo-political bonanza is a temporary relief that is not sustainable . Education and economic development seem miles away”**(Presupposition/Metaphor/Compassion)**
(iii)	8616	0	0	1	”We want all poor families to earn enough and save enough . The rural savings banks were meant to encourage that” **(borderline rhetoric)**
(iii)	8838	0	1	1	“I always consider this job as a gift , being a nurse is a reward and task given by God to help those who are in need . Seeing your patient recover from an illness , watching their families smile when you give them care , and hearing the first cry of a newborn are just some of the things that make my work special . It might be a heavy work but it can lighten your heart, she expressed.” **(positive sentiment mistaken for PCL)**
(iv)	157	1	0	0	“We are alarmed to learn of your recently circulated proposals that would eviscerate the Lifeline program and leave many of the most vulnerable people in the country without access to affordable communications . As you are well aware , the Lifeline program provides a modest monthly subsidy of $9.25 to connect low-income Americans to phone and internet services . As broadband prices continue to soar , and affordability continues to suffer , adoption gaps remain . The Lifeline program has proven critical for poor families and people of color who are caught on the wrong side of the digital divide .” **(strong positive (4); missed by both)**

Table 15: Representative error analysis examples on the official dev set.

scension (where the patronising stance is pragmatic rather than explicit), and (ii) **borderline rhetoric** where supportive/compassionate language resembles PCL.

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